

MARGVEDHA: Critical Technical Assessment and Implementation Blueprint

****Prepared for:**** MARGVEDHA Hackathon Team

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****Status:**** Research Dossier — All claims verified against available sources

1. Executive Summary

MARGVEDHA proposes to build a dependency vulnerability propagation analyzer with interactive graph visualization and evidence-based remediation planning. The core problem statement—CSB-03: Supply-Chain Vulnerability Propagation Graph—is technically valid and addresses a real gap in how security teams understand transitive vulnerability exposure.

****Critical finding:**** The SCA tooling landscape is mature. OSV-Scanner, Snyk, Dependabot, Dependency-Track, and others already handle dependency parsing, OSV queries, and basic transitive analysis. A tool that merely "scans manifests and shows vulnerable dependencies" is redundant.

****The defensible contribution space**** lies in the intersection of four capabilities that are currently fragmented across tools:

1. ****Explanation-first graph traversal**** — showing not just which projects are affected, but ***why*** (the precise dependency path with intermediate packages labeled).
2. ****Fix-impact simulation**** — before-and-after graph states for candidate upgrades, using resolver-validated version changes where possible.

3. **Multi-project batch analysis with shared-dependency clustering** — identifying common vulnerable packages across uploaded manifests and optimizing patch order for maximum finding reduction.
4. **Evidence-tiered risk scoring** — explicitly distinguishing advisory matches, potential exposure, and any verified reachability, without conflating them.

The recommended MVP for a student hackathon prioritizes **static manifest parsing → OSV batch querying → graph construction → explanation rendering → simulated fix comparison** over speculative reachability analysis. Reachability is deferred because it requires language-specific static analysis that is impractical to build reliably within hackathon constraints.

Claims that are safe: "MARGVEDHA maps transitive vulnerability propagation with explainable paths and simulates remediation impact."

Claims that are not supported: "First to combine graph analysis with remediation planning" or "reduces false positives through reachability."

2. Problem-Statement Interpretation

2.1 What CSB-03 Actually Requires

The official brief specifies four technical capabilities:

Requirement	Technical Interpretation	Hard Parts
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Parse sample manifests (`package.json` , `requirements.txt` , `pom.xml`) Extract declared dependencies with version constraints Manifests alone do not contain transitive dependencies
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| Build dependency graph | Represent project→package→package relationships | Resolution requires lockfiles or ecosystem-specific resolvers |

| Highlight directly and transitively affected projects | Traverse graph from vulnerable nodes backward to roots | Multiple vulnerable versions, multiple root projects, deduplication |

| Rank by exploitability/exposure | Multi-factor scoring using CVSS, KEV, EPSS, topology | No single "correct" formula; missing data handling |

| Suggest patch order | Prioritize fixes that resolve most findings | Compatibility constraints are not visible from OSV alone |

2.2 Implicit Requirements Not Stated in the Brief

The brief says "given sample manifests" but does not specify:

- **Lockfile availability:** Whether the test inputs include lockfiles (which contain resolved versions and transitive dependencies) or only manifests (which contain version ranges). This is the single most important determinant of graph completeness.

- **Scope of "projects":** Whether MARGVEDHA analyzes a single project with multiple manifests, multiple projects uploaded separately, or a monorepo with cross-dependencies.

- **Remediation validation:** Whether a "suggested fix" must be resolver-validated (a version combination that actually satisfies all constraints) or merely advisory-based (a version range from OSV's `fixed` field).

Recommendation: Interpret "sample manifests" as **manifests plus lockfiles when available**, and explicitly report when only a manifest is provided (limiting transitive analysis to declared dependencies).

2.3 The Five Distinct States (Critical Distinction)

The prompt asks to explain why these are different. This distinction is the foundation of responsible risk reporting:

State	Evidence Required	What It Means	MARGVEDHA Capability
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1. Advisory match	OSV returns a vulnerability for a package@version	The specific version is listed as affected by a known advisory	Fully supported
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2. Project includes package	Package appears in dependency tree	The vulnerable code is present in the dependency closure	Fully supported (if lockfile available)
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3. Reachable through dependency path	Path exists from root project to vulnerable package	There is a structural path, but runtime execution is not proven	Fully supported (path visualization)
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4. Vulnerable code reachable at runtime	Static call-graph analysis shows the vulnerable function is invoked	The specific vulnerable code path can execute	Not supported in MVP
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5. Application is exploitable	Runtime conditions, input vectors, configuration all align	An attacker can actually trigger the vulnerability	Not claimable
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Why this matters: Most SCA tools—including OSV-Scanner and Dependabot—stop at states 1–2. Snyk reaches state 4 for Java (Maven/Gradle) with mature call-graph analysis, and state 3 for JavaScript/TypeScript with limitations. No tool reliably reaches state 5 without manual verification.

MARGVEDHA must label its outputs with the highest state it has actually established.

3. Technical Foundations

3.1 Glossary for Engineering Students

| Term | Definition | Relevance to MARGVEDHA |

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| **Software Supply Chain** | The ecosystem of dependencies, build tools, and distribution channels that produce a software artifact | The domain problem |

| **SCA (Software Composition Analysis)** | Tooling that inventories third-party components and matches them against vulnerability databases | What MARGVEDHA is, broadly |

| **CVE** | Common Vulnerabilities and Exposures—a unique identifier for a publicly disclosed vulnerability | The "what" being tracked |

| **CWE** | Common Weakness Enumeration—a category of software weakness (e.g., CWE-79: XSS) | Classification for grouping |

| **CVSS** | Common Vulnerability Scoring System—a severity score (0–10) based on exploitability and impact metrics | One input to risk scoring, not a probability |

| **CPE** | Common Platform Enumeration—a structured naming scheme for IT products (`cpe:2.3:a:vendor:product:version``) | NVD uses CPE; OSV uses PURL—matching challenge |

| **PURL** | Package URL—a standardized identifier for software packages (`pkg:npm/lodash@4.17.20``) | OSV's primary package identity format |

| **SBOM** | Software Bill of Materials—a machine-readable inventory of components in a software artifact | Optional input/output; not required by the brief |

| **Transitive Dependency** | A dependency of a dependency (depth ≥ 2 from the project root) | The core propagation problem |

| **Version Constraint** | A range expression (e.g., `^1.2.3``) specifying acceptable versions | Manifests contain constraints; lockfiles contain resolved versions |

| **Lockfile** | A file recording the exact resolved versions of all dependencies | Required for complete transitive graph construction |

| **Vulnerability Matching** | Determining whether a specific package@version is affected by an advisory | OSV API handles this; the challenge is query efficiency and correctness |

| **Runtime Reachability** | Whether the vulnerable function is actually invoked during execution | Not determinable from manifests alone |

| **EPSS** | Exploit Prediction Scoring System—a probability (0–1) that a vulnerability will be exploited in the next 30 days | Available via FIRST.org API |

| **KEV** | Known Exploited Vulnerabilities—CISA's catalog of vulnerabilities confirmed exploited in the wild | Binary signal: present or absent |

| **Fix Availability** | Whether a patched version exists and is identified in the advisory | OSV advisories include `fixed` version ranges |

| **Patch-Order Optimization** | Sequencing fixes to maximize risk reduction per unit of remediation effort | MARGVEDHA's proposed contribution area |

3.2 The Five States: Detailed Explanation

State 1 — Advisory Match:

OSV receives `package=npm:lodash, version=4.17.19` and returns advisory GHSA-xxxx with `affected: [{"package": {"ecosystem": "npm", "name": "lodash"}, "ranges": [{"type": "SEMVER", "events": [{"introduced": "0"}, {"fixed": "4.17.20"}]}]}`. This proves only that the version appears in an advisory's affected range.

State 2 — Project Includes Package:

The project's dependency tree contains lodash@4.17.19. If the project uses `package-lock.json`, this is resolved unambiguously. If it only has `package.json` with `lodash: "^4.17.0"`, the actual installed version is unknown without resolution.

State 3 — Reachable Through Dependency Path:

A path exists: `my-app → express@4.18.2 → body-parser@1.20.1 → qs@6.11.0`. If qs@6.11.0 has a vulnerability, the path proves the code is in the dependency closure. It does not prove the vulnerable function in qs is ever called.

State 4 — Vulnerable Code Reachable at Runtime:

Snyk's reachability analysis for Java builds a call graph from application entry points, through dependencies, to the specific method identified in the advisory. This requires

source code, not just manifests. For JavaScript/Python, dynamic features (dynamic imports, monkey-patching, duck typing) make static reachability unreliable .

****State 5 — Application Exploitable:****

Requires runtime context: is the vulnerable function exposed to attacker-controlled input? Is the vulnerable feature enabled? Are there mitigating controls? No static tool can determine this; it requires manual verification or runtime instrumentation.

****MARGVEDHA should label every finding with the highest state established, never implying a higher state.****

4. Current Data Sources and Integration

4.1 OSV.dev

****Purpose:**** Aggregates vulnerability data from ecosystem-native sources (GitHub Advisory, PyPI Advisory, RustSec, Go vulndb, etc.) into a single schema with consistent version-range semantics .

****Key facts:****

- Free, no API key required for public tier
- Update lag: < 24 hours for major ecosystems
- Supports per-package queries (` POST /v1/query`) and batch queries (` POST /v1/querybatch` , up to 1000 per request)
- Also provides daily snapshots via public GCS bucket

****Critical schema gotchas:****

- `affected` field can contain multiple ranges; comparator depends on ecosystem (SemVer for npm/Go, PEP 440 for PyPI, Maven ordering)
- Withdrawn vulnerabilities remain in the database; filter on `withdrawn` timestamp
- `severity` is not always populated; when present, scoring system varies (CVSS v3 or v4)
- Aliases (CVE, GHSA) must be deduplicated by the consumer

****Example query (verified from documentation):****

```http

POST https://api.osv.dev/v1/query

Content-Type: application/json

```
{
 "package": {
 "name": "lodash",
 "ecosystem": "npm"
 },
 "version": "4.17.19"
}
```

```

****Integration recommendation:**** Use `/v1/querybatch` for all package@version pairs extracted from the project, then fetch full advisory details only for the subset that returned vulnerability IDs. This pattern resolves a 2000-package SBOM in two batch calls plus 10–40 detail fetches .

4.2 CISA Known Exploited Vulnerabilities (KEV) Catalog

****Purpose:**** Curated list of vulnerabilities confirmed to be actively exploited in the wild. Presence is a binary signal of real-world exploitation .

****Key facts:****

- Available as JSON/CSV download; no API key required
- Fields include: CVE ID, vendor/project, product, vulnerability name, date added, due date, known ransomware use
- For U.S. federal agencies, KEV membership carries a binding remediation deadline

****Integration recommendation:**** Load the KEV catalog at scan time (it's small, ~1000–1500 entries) and cross-reference by CVE alias. A KEV match should elevate priority regardless of CVSS score.

4.3 NVD and CVSS

****Purpose:**** The authoritative CVE feed with CVSS scores, CWE classifications, and CPE identifiers.

****Critical limitation:**** NVD has been operationally unreliable since an analysis backlog began in early 2024 and has not fully recovered . Update delay is 5–14 days .

****When to use:**** For CVSS scores when OSV does not provide them, and for CWE classification. Do not use NVD as the primary vulnerability source for package-version matching; OSV is faster and more accurate for this purpose.

4.4 GitHub Advisory Database

****Purpose:**** Curated advisories from GitHub, often published days before NVD enrichment .

****Key facts:****

- Available via OSV.dev (GitHub Advisory is an OSV source)
- Fields: GHSA ID, CVE alias, severity, CVSS score/vector, CWE IDs, affected packages with version ranges, first patched version
- One advisory can cover multiple packages (e.g., lodash, lodash-es, lodash.template)

****Integration recommendation:**** OSV already aggregates GitHub Advisory. No separate integration needed.

4.5 EPSS

****Purpose:**** Probability (0–1) that a vulnerability will be exploited in the next 30 days, based on real-world exploitation data .

****Key facts:****

- Available via FIRST.org API: `https://api.first.org/data/v1/epss?cve=CVE-2024-27198``
- Also available as daily CSV/JSON downloads
- Scores most CVEs low; a small fraction score high (this is the signal)

****Integration recommendation:**** Optional enrichment. If EPSS API is unavailable, degrade gracefully: KEV presence and CVSS severity remain usable signals. Document that EPSS is unavailable.

4.6 Minimum Viable Data Strategy

| Signal | Source | Priority | Fallback if Unavailable |

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Advisory match (package@version)	OSV API	**Required**	Cannot function
Fixed version	OSV advisory	**Required**	Report "no fix available"
CVSS severity	OSV `severity` or NVD	**Recommended**	Default to "unknown"
CVE alias	OSV `aliases`	**Recommended**	Use OSV ID
KEV membership	CISA KEV (cached)	**Optional**	Report "KEV status unavailable"
EPSS probability	FIRST.org API	**Optional**	Omit from scoring
Runtime reachability	Static analysis	**Deferred**	Never claim

5. Competitor and Market Analysis

5.1 Feature Comparison Table

Tool	Input Formats	Transitive Analysis	Graph Visualization	Reachability	Risk Prioritization	Guided Remediation	Cost
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OSV-Scanner	Manifests, lockfiles, containers, binaries	Yes (via OSV-SCALIBR)	Interactive HTML (v2)	Planned (not shipped)	Severity filtering	Guided remediation for npm, Maven	Free
Snyk Open Source	Manifests, lockfiles, SCM integration	Yes	Dependency tree view	Java (mature), JS/TS (limited), Python (limited)	Multi-factor (reachability + EPSS/KEV + asset criticality)	Automated fix PRs	Free tier + paid
GitHub Dependabot	Manifests, lockfiles (14+ ecosystems)	Yes (via dependency graph)	Dependency graph view	No	Severity-based	Automated PRs	Free
OWASP Dependency-Track	SBOM (CycloneDX)	Yes	Portfolio view	No	Multi-factor (CVSS, EPSS)	No	Free

Socket.dev	npm, PyPI	Yes	No	Yes (JS-specific)	Risk score	Yes	Free tier + paid
Trivy	Container images, filesystems, manifests	Yes	No	No	Severity	No	Free
Syft + Gype	SBOM generation + scanning	Yes	No	No	Severity	No	Free

Legend:

- Confirmed feature
- Feature not confirmed from available documentation
- Requires specific plan/configuration

5.2 What Is Already Standard

The following capabilities are table stakes and cannot serve as MARGVEDHA's novelty claim:

- Parsing manifests for npm, PyPI, Maven
- Querying OSV for package@version matches
- Basic dependency tree rendering
- CVSS-based severity ranking
- SBOM export (CycloneDX, SPDX)

5.3 Where Gaps Remain

Gap	Evidence	MARGVEDHA Opportunity
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Explainable propagation paths	OSV-Scanner shows findings but not the precise path from root project to vulnerable package with intermediate labels	Path visualization with human-readable traversal
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| **Multi-project shared-dependency clustering** | No mainstream free tool clusters common vulnerable packages across uploaded projects | Batch analysis feature |

| **Fix-impact simulation with resolver awareness** | OSV-Scanner guided remediation modifies files; it does not show a simulated before/after graph | What-if upgrade visualization |

| **Evidence-tiered labeling** | Tools conflate "present" with "potentially exploitable" | Explicit state labeling (advisory match vs. path reachability) |

| **Educational/developer-facing explanations** | Commercial tools focus on enterprise workflows | Explain *why* a path matters, not just *that* it exists |

5.4 Features to Avoid Claiming as Novel

- "First to use OSV for vulnerability detection" — OSV-Scanner exists
- "First dependency graph visualization" — Dependabot, Snyk, and others have this
- "First to support transitive vulnerabilities" — Standard in modern SCA
- "First to suggest patch order" — OSV-Scanner's guided remediation does this for npm and Maven

6. Verified Literature Review

Note on methodology: The following literature was identified through search results. For papers where only abstracts are available, limitations are inferred from the abstract and known research context. Where publication venue is not stated in search results, it is marked as unverified.

6.1 Core Papers on Vulnerability Propagation

****Paper 1: VDGraph: A Graph-Theoretic Approach to Unlock Insights from SBOM and SCA Data****

- ****Authors:**** Howell Xia, Şevval Şimşek, et al.
- ****Year:**** 2025
- ****Venue:**** arXiv (cs.SE)
- ****DOI/URL:**** arXiv:2507.20502
- ****Research Question:**** How to integrate SBOM and SCA outputs into a unified dependency-vulnerability graph for analysis
- ****Method:**** Knowledge graph integrating CycloneDX Maven plugin SBOM and OSV-Scanner output into Neo4j; formal graph properties and conflict resolution
- ****Dataset:**** 21 popular Java projects
- ****Key Findings:**** Vulnerabilities predominantly emerge at depth ≥ 3 dependency levels; direct dependencies exhibit lower vulnerability density; concentrated risk points (vulnerable components reachable through numerous paths)
- ****Limitations:**** Java-only; proof-of-concept scale
- ****Relevance to MARGVEDHA:**** Directly informs graph schema design and path analysis algorithms
- ****Design Decision Informed:**** Node types (project, package-version, vulnerability) and edge semantics (depends-on, affects)

****Paper 2: Vulnerability Propagation Analysis (VPSS)****

- ****Title (inferred):**** "An Accurate and Efficient Vulnerability Propagation Analysis Framework"
- ****Authors:**** Not available from search result
- ****Year:**** 2025 (arXiv:2506.01342)
- ****Venue:**** arXiv
- ****URL:**** ar5iv.labs.arxiv.org/html/2506.01342
- ****Research Question:**** How to quantify vulnerability impact in software supply chains with time-awareness

- **Method:** Vulnerability Propagation Scoring System (VPSS), time-aware indicator; call-graph level analysis for Maven
- **Dataset:** Real-world Java Maven vulnerabilities
- **Key Findings:** Distinguishes PV-level (package-version declared) from CG-level (call graph) propagation; call-graph analysis reduces overestimation of affected projects
- **Limitations:** Maven ecosystem only; prototype
- **Relevance:** Validates the distinction between "declared dependency" and "actual code reachability"
- **Design Decision:** MARGVEDHA should report both levels but only claim PV-level analysis

Paper 3: Tracing Vulnerabilities in Maven

- **Title:** "Tracing Vulnerabilities in Maven: A Study of CVE lifecycles and Dependency Networks"
- **Authors:** Not available from search result
- **Year:** 2025
- **Venue:** arXiv (2502.04621)
- **URL:** export.arxiv.org/pdf/2502.04621
- **Research Question:** What is the lifecycle of vulnerabilities in Maven artifacts?
- **Method:** Analysis of 175 repositories, 456 CVEs; dependency patch adoption behavior
- **Key Findings:** 81.6% of CVEs in Maven are patch-before-publish; reactive adoption averages 78 days for critical severity; maintainers prioritize critical patches faster than lower severity
- **Limitations:** Maven-specific; correlation analysis
- **Relevance:** Informs remediation urgency estimates
- **Design Decision:** Patch-order algorithm should prioritize critical-severity CVEs with available fixes

6.2 Papers on Reachability and SCA Limitations

****Paper 4: Snyk Reachability Analysis Mechanics****

- ****Title:**** "How Snyk's Reachability Analysis Determines Whether Vulnerable Code Is Actually Called"
- ****Authors:**** Safeguard (industry blog)
- ****Year:**** 2026
- ****Venue:**** Safeguard.sh (industry analysis)
- ****URL:**** safeguard.sh/resources/blog
- ****Research Question:**** How does Snyk's static call-graph reachability work?
- ****Key Findings:**** Java has deepest coverage (static typing, explicit imports); JavaScript/Python have structural limitations (dynamic imports, monkey-patching); verdicts are discrete (reachable / no path found / unsupported)
- ****Relevance:**** MARGVEDHA should not claim reachability without implementing call-graph analysis
- ****Design Decision:**** Defer reachability; label findings as "path-reachable" not "runtime-reachable"

****Paper 5: SCA Tool Limitations in Maven****

- ****Title:**** "Measuring the Security of the Evolving Software Supply Chain: a Research Agenda"
- ****Author:**** Sarah Meriem Ourari
- ****Year:**** 2026
- ****Venue:**** arXiv (2609.08810)
- ****URL:**** export.arxiv.org/pdf/2609.08810
- ****Research Question:**** What are the gaps in measuring software supply chain security?

- **Key Findings:** SCA tools focus on direct dependencies, ignore resolution issues, produce false positives; "no distinction between types of external libraries"; need for Scan Scope Maven concept
- **Relevance:** Validates the need for careful manifest parsing and explicit limitation reporting
- **Design Decision:** MARGVEDHA should distinguish `dependencies`, `devDependencies`, `optionalDependencies` in parsing

6.3 Papers on Risk Prioritization

Paper 6: EPSS (Exploit Prediction Scoring System)

- **Title:** "Exploit Prediction Scoring System"
- **Authors:** Jay Jacobs, Sasha Romanosky, Benjamin Edwards, Michael Roytman, Idris Adjerid
- **Year:** 2021
- **Venue:** ACM Digital Threats: Research and Practice, 2(3)
- **DOI:** 10.1093/cybsec/tyaa015
- **URL:** first.org/resources/papers/conf2023
- **Research Question:** Can exploit probability be predicted from vulnerability metadata?
- **Method:** Machine learning on historical exploitation data
- **Key Findings:** EPSS outperforms CVSS for predicting exploitation; freely available for 180,000+ vulnerabilities
- **Relevance:** Core input to MARGVEDHA's risk scoring
- **Design Decision:** EPSS is optional enrichment; degrade to KEV + CVSS if unavailable

Paper 7: Risk-Aware Prioritization Approaches

- **Source:** IEEE paper (stamp.jsp?arnumber=11393471)
- **Year:** 2025

- **URL:** ieeexplore.ieee.org/stamp
- **Key Findings:** Combining EPSS, KEV, and reachability improves Precision@K and reduces MTTR vs. severity-only queues; reachability-aware SCA down-ranks dormant CVEs
- **Relevance:** Justifies multi-factor scoring
- **Design Decision:** MARGVEDHA's risk score combines CVSS, KEV, EPSS (if available), and graph topology (direct vs. transitive)

6.4 Papers on SBOM and Ecosystem Security

Paper 8: SBOM Dataset and Tool Benchmarking

- **Title:** "Assessing the Security of Software Supply Chains: Software Bill of Materials, Threat Propagation, and Logical Attack Graphs"
- **Author:** Luís Fernando de Oliveira Soeiro
- **Year:** 2025
- **Venue:** Institut Polytechnique de Paris (PhD thesis)
- **URL:** theses.hal.science/tel-05516576
- **Key Findings:** 78,000+ unique SBOMs deduplicated from 94M+ repositories; most SBOMs not adequately prepared for use; significant disagreement among tools; current SCA tools fail to account for SSC interactions
- **Relevance:** Validates the problem space; warns against over-reliance on any single tool
- **Design Decision:** MARGVEDHA should report uncertainty when data is incomplete

Paper 9: Cargo Dependency-Vulnerability Knowledge Graph

- **Title:** "Cargo Ecosystem Dependency-Vulnerability Knowledge Graph Construction and Vulnerability Propagation Study"
- **Authors:** Not available from search result
- **Year:** 2022

- **Venue:** arXiv (2210.07482)
- **URL:** arxiv.org/html/2210.07482
- **Method:** Neo4j knowledge graph from crates.io (75,922 libraries) and GitHub Advisory (351 CVEs)
- **Key Findings:** Static analysis without official resolution rules produces low accuracy; considering official resolution rules (dev dependencies, optional, features) improves correctness
- **Relevance:** Validates the need for ecosystem-aware parsing
- **Design Decision:** MARGVEDHA should document which dependency types are included

6.5 Literature Synthesis

Common findings across papers:

1. Manifest-only analysis produces incomplete graphs; lockfiles or resolution rules are required
2. Package-level presence overestimates affected projects; call-graph analysis reduces false positives
3. CVSS alone is insufficient for prioritization; EPSS, KEV, and reachability improve ranking
4. Vulnerabilities cluster at depth ≥ 3 ; direct dependencies are less vulnerable
5. Multi-factor risk scoring is standard in research and industry

Unresolved problems:

- Cross-ecosystem unified dependency modeling remains fragmented
- Reachability analysis quality varies dramatically by language
- No standard benchmark for evaluating SCA tool accuracy across ecosystems

7. Research Gaps and Novelty Assessment

7.1 Proposed Feature Evaluation

Feature	Existing Implementations	Genuine Gap?	User Value	Implementation Complexity	Hackathon Suitable?
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A. Explainable propagation graph	Dependabot (basic tree), Snyk (tree view), OSV-Scanner (interactive HTML)	Presentation gap, not fundamental	High for debugging	Medium (graph traversal + UI)	**Yes**
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B. Multi-finding remediation planner	OSV-Scanner guided remediation (npm, Maven), Snyk fix PRs	Gap in free, non-commercial tools	High for triage	High (requires resolver integration)	**Partial** (simulate, don't validate)
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C. Before/after graph simulation	Not found in free tools	Genuine gap	High for decision-making	Medium (duplicate graph, apply changes)	**Yes**
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D. Topology-aware risk ranking	Snyk Risk Score, Kusari Score	Gap in open tools; commercial exists	Medium	Low (scoring formula)	**Yes**
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E. Evidence-tiered labeling	Not standard in free tools	Genuine gap	High for honesty/trust	Low (explicit state labels)	**Yes**
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F. Scan comparison (resolved/new)	OSV-Scanner has basic diff	Minor gap	Medium	Low (set operations)	**Yes**
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G. Cross-project clustering	Dependency-Track (portfolio view)	Gap in lightweight tools	High for orgs	Medium (index packages across projects)	**Yes**
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H. Uncertainty-aware reporting	Rare	Genuine gap	Medium	Low (report missing data)	**Yes**
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I. Patch-order optimization	OSV-Scanner guided remediation	Gap in multi-project context	Medium-High	High (compatibility constraints)	**Partial**
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7.2 Recommended Contributions

****Primary Innovation: Explainable Multi-Project Propagation Graph with Fix-Impact Simulation****

This combines features A, C, and G into a coherent capability:

- Upload multiple manifests/lockfiles
- Build a unified graph across all projects and dependencies
- Identify vulnerable packages and traverse paths back to root projects
- Render each path with intermediate package labels and depth
- Simulate a candidate upgrade (e.g., "upgrade lodash to 4.17.21") and show:
 - Which findings are resolved
 - Which projects are affected
 - Any new dependency paths introduced
- Rank projects by shared-vulnerability clustering

****Why this is defensible:**** No free tool combines multi-project batch analysis, explainable path rendering, and simulated fix impact in a single interface. Each component exists elsewhere, but the integration for developer triage is the contribution.

****Supporting Innovation 1: Evidence-Tiered Risk Scoring****

Every finding is labeled with the highest state established:

- `ADVISORY\_MATCH` : package@version appears in advisory
- `PROJECT\_CONTAINS` : package is in dependency tree
- `PATH\_REACHABLE` : dependency path exists from root to vulnerable package
- (`RUNTIME\_REACHABLE` : not claimed)

Risk score combines: CVSS severity + KEV membership + EPSS (if available) + topology (direct/transitive) + fix availability. The scoring formula is exposed to the user.

****Supporting Innovation 2: Multi-Project Patch-Order Suggestion****

Given multiple projects and findings, suggest an ordered fix list that:

1. Prioritizes fixes that resolve findings across multiple projects
2. Prioritizes KEV vulnerabilities
3. Prefers upgrades with available fixed versions
4. Reports when a fix requires a major version change (compatibility risk flag)

This is advisory only; MARGVEDHA does not modify files.

7.3 Classification of Contributions

Contribution Classification

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Multi-project propagation graph Engineering integration + UX improvement
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Fix-impact simulation UX improvement + engineering
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Evidence-tiered labeling UX improvement (transparency)
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Multi-project patch ordering Engineering integration
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Reachability analysis **Deferred** — would require research contribution
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8. Graph Model and Algorithms

8.1 Graph Schema

****Directed graph**** `G = (V, E)` where:

****Node types:****

Node Type	Attributes	Example
`Project`	`id`, `name`, `source_file`	project:my-app
`PackageVersion`	`id`, `ecosystem`, `name`, `version`	pkg:npm/lodash@4.17.19
`Vulnerability`	`id`, `aliases`, `severity`, `fixed_versions`	GHSA-xxxx

****Edge types:****

Edge Type	From → To	Semantics
`DEPENDS_ON`	Project → PackageVersion	Project declares/resolves this dependency
`DEPENDS_ON`	PackageVersion → PackageVersion	Transitive dependency
`AFFECTS`	Vulnerability → PackageVersion	Advisory applies to this version
`ALIAS_OF`	Vulnerability → Vulnerability	Cross-reference (CVE ↔ GHSA)

****Edge attributes for `DEPENDS\_ON`:****

- `dependency\_type`: `direct` | `transitive`
- `declared\_range`: version constraint (e.g., `^4.17.0`)
- `optional`: boolean

- `dev` : boolean

8.2 Example Graph (JSON)

```
` `` `json
{
  "nodes": [
    {"id": "project:web-app", "type": "Project", "name": "web-app"},
    {"id": "pkg:npm/express@4.18.2", "type": "PackageVersion", "ecosystem": "npm", "name":
"express", "version": "4.18.2"},
    {"id": "pkg:npm/body-parser@1.20.1", "type": "PackageVersion", "ecosystem": "npm",
"name": "body-parser", "version": "1.20.1"},
    {"id": "pkg:npm/qs@6.11.0", "type": "PackageVersion", "ecosystem": "npm", "name": "qs",
"version": "6.11.0"},
    {"id": "GHSA-xxxx-yyyy-zzzz", "type": "Vulnerability", "aliases": ["CVE-2022-24999"],
"severity": 7.5, "fixed_versions": ["6.11.1"]}
  ],
  "edges": [
    {"from": "project:web-app", "to": "pkg:npm/express@4.18.2", "type": "DEPENDS_ON",
"dependency_type": "direct"},
    {"from": "pkg:npm/express@4.18.2", "to": "pkg:npm/body-parser@1.20.1", "type":
"DEPENDS_ON", "dependency_type": "transitive"},
    {"from": "pkg:npm/body-parser@1.20.1", "to": "pkg:npm/qs@6.11.0", "type":
"DEPENDS_ON", "dependency_type": "transitive"},
    {"from": "GHSA-xxxx-yyyy-zzzz", "to": "pkg:npm/qs@6.11.0", "type": "AFFECTS"}
  ]
}
` `` `
```

8.3 Traversal Pseudocode

****Find affected projects:****

```
```python
```

```
def find_affected_projects(graph, vulnerable_package_ids):
```

```
 """
```

```
 Given a set of vulnerable package version node IDs,
```

```
 find all root Project nodes that have a dependency path to them.
```

```
 Returns a dict: project_id -> list of paths
```

```
 """
```

```
 affected = {}
```

```
 for vuln_pkg_id in vulnerable_package_ids:
```

```
 # BFS backward from vulnerable package to root projects
```

```
 visited = set()
```

```
 queue = [(vuln_pkg_id, [vuln_pkg_id])]
```

```
 while queue:
```

```
 current, path = queue.pop(0)
```

```
 if current in visited:
```

```
 continue
```

```
 visited.add(current)
```

```
 for pred in graph.predecessors(current):
```

```

 if graph.nodes[pred]["type"] == "Project":
 if pred not in affected:
 affected[pred] = []
 affected[pred].append(list(reversed(path + [pred])))
 else:
 queue.append((pred, path + [pred]))

 return affected
'''

'''Find shortest paths:'''

'''python
def shortest_paths_to_vuln(graph, project_id, vuln_pkg_id):
 """Return all shortest dependency paths from project to vulnerable package."""
 return list(nx.all_shortest_paths(graph, source=project_id, target=vuln_pkg_id))
'''

```

### ### 8.4 Graph Centrality: Why It Does Not Prove Exploitability

Betweenness centrality or PageRank on the dependency graph identifies packages that are "structurally important" — many paths pass through them. This is useful for identifying shared dependencies but does **not** indicate:

- Whether the vulnerable function is called
- Whether the vulnerability is exploitable in this configuration
- Whether the package version in this project is actually affected

**\*\*MARGVEDHA must not claim that centrality equals risk.\*\*** Centrality is a topology signal; risk requires exploitability evidence.

### ### 8.5 Visualization Library Recommendation

**\*\*Cytoscape.js\*\*** is recommended:

- Mature, actively maintained
- Rich layout algorithms (dagre for dependency trees, cose for clustered graphs)
- Interactive: zoom, pan, node click for details
- Works well with React via `react-cytoscapejs`
- Handles graphs with thousands of nodes performantly with proper configuration

**\*\*Alternative:\*\*** `react-flow` (simpler API, better for small-medium graphs, less mature for complex layouts).

**\*\*Recommendation:\*\*** Cytoscape.js with dagre layout for dependency trees; custom node styling for project vs. package vs. vulnerability distinction.

## ## 9. Dependency-Resolution Strategy

### ### 9.1 Ecosystem-by-Ecosystem Analysis

**\*\*JavaScript (npm):\*\***

- **\*\*Manifest:\*\*** `package.json` — contains `dependencies`, `devDependencies`, `optionalDependencies` with version ranges (e.g., `^4.17.0`)

- **Lockfile:** `package-lock.json` — contains resolved versions and full transitive tree; required for complete graph
- **Parsing library:** `@npmcli/package-json` or simple JSON parsing
- **Resolution:** If lockfile present, extract `packages` field (v2/v3 format) or `dependencies` (v1 format); if only manifest, transitive analysis is impossible without running `npm install` (not recommended)
- **OSV identity:** `pkg:npm/<name>@<version>`

## **Python:**

- **Manifest:** `requirements.txt` — may contain exact pins (`==`) or ranges (`>=`, `~=`); often no transitive info
- **Lockfile:** `poetry.lock` (Poetry), `Pipfile.lock` (Pipenv) — contain resolved versions
- **Parsing library:** `packaging` for version comparison; `pip-requirements-parser` for `requirements.txt`
- **Resolution:** If lockfile present, extract resolved packages; if only `requirements.txt` with ranges, transitive graph is incomplete
- **Name normalization:** PyPI is case-insensitive, hyphens/underscores equivalent (`my-package` == `my_package`); normalize to lowercase with hyphens
- **OSV identity:** `pkg:pypi/<normalized_name>@<version>`

## **Java (Maven):**

- **Manifest:** `pom.xml` — contains `<dependencies>` with `<version>` (may be ranges or properties)
- **Resolution:** Maven's dependency mediation is complex (nearest-wins, `dependencyManagement` overrides); `pom.xml` alone is insufficient for transitive graph
- **Tool:** Maven itself can generate dependency tree (`mvn dependency:tree`), but this requires running Maven (execution risk)
- **Recommended approach:** If `pom.xml` only, parse direct dependencies and report that transitive analysis is limited. If `pom.xml` plus a lockfile (rare in Maven), use that.

- **OSV identity:** `pkg:maven/<groupId>/<artifactId>@<version>`

### 9.2 MVP Strategy

**Recommended input handling:**

Input Provided	Graph Completeness	MARGVEDHA Behavior
----------------	--------------------	--------------------

-----	-----	-----
-------	-------	-------

Manifest + Lockfile	Full transitive graph	Complete analysis
---------------------	-----------------------	-------------------

Manifest only (npm/PyPI)	Direct dependencies only	Report "transitive analysis limited—no lockfile"
--------------------------	--------------------------	--------------------------------------------------

Manifest only (Maven)	Direct dependencies (with version mediation caveats)	Report "transitive analysis unavailable—pom.xml only"
-----------------------	------------------------------------------------------	-------------------------------------------------------

Lockfile only	Full transitive graph	Parse lockfile; infer project name from lockfile metadata
---------------	-----------------------	-----------------------------------------------------------

**Do not claim complete transitive analysis when only manifests are provided.**

### 9.3 Package Identity Normalization

Ecosystem	Normalization	OSV Field
-----------	---------------	-----------

-----	-----	-----
-------	-------	-------

npm	Lowercase; scoped packages retain `@scope/name`	`ecosystem: "npm"`, `name: "lodash"` or `name: "@babel/core"`
-----	-------------------------------------------------	------------------------------------------------------------------

PyPI	Lowercase; hyphens/underscores → hyphens	`ecosystem: "PyPI"`, `name: "django"`
------	------------------------------------------	---------------------------------------

Maven	Preserve case; `groupId:artifactId`	`ecosystem: "Maven"`, `name: "org.apache.commons:commons-text"`
-------	-------------------------------------	-----------------------------------------------------------------

## ## 10. Explainable Risk Scoring and Patch Planning

### ### 10.1 Risk Scoring Model

**\*\*Formula (configurable weights):\*\***

...

```
risk_score = w_cvss * cvss_normalized
 + w_kev * kev_present
 + w_epss * epss_score (if available)
 + w_topology * topology_factor
 + w_fix * fix_availability
 + w_shared * shared_dependency_count_normalized
```

...

**\*\*Factor definitions:\*\***

Factor	Normalization	Weight (default)	Notes
--------	---------------	------------------	-------

CVSS severity	<code>`cvss_base` / 10.0</code>	0.30	If missing, use 0.0 and flag
---------------	---------------------------------	------	------------------------------

KEV presence	1.0 if in KEV, 0.0 otherwise	0.35	Strong signal
--------------	------------------------------	------	---------------

EPSS probability	<code>`epss`</code> (already 0–1)	0.15	Optional; if unavailable, distribute weight to CVSS/KEV
------------------	-----------------------------------	------	---------------------------------------------------------

Topology (direct vs. transitive)	1.0 direct, 0.6 transitive	0.10	Direct dependencies are more visible to developers
----------------------------------	----------------------------	------	----------------------------------------------------

| Fix availability | 1.0 if fixed version exists, 0.0 otherwise | 0.05 | Unfixable findings are less actionable |

| Shared dependency count |  $\min(1.0, \text{count} / 5)$  | 0.05 | A vulnerability affecting 5+ projects is higher priority |

#### **\*\*Missing data handling:\*\***

- If EPSS unavailable: redistribute EPSS weight equally to CVSS and KEV
- If CVSS missing: use 0.5 (neutral) and flag "severity unknown"
- If fix unavailable: score remains, but finding is labeled "no fix available"

#### **\*\*Score interpretation:\*\***

- 0.0–0.3: Low priority (backlog)
- 0.3–0.6: Medium priority (next release)
- 0.6–0.8: High priority (this sprint)
- 0.8–1.0: Critical priority (act now)

### ### 10.2 Worked Example (Illustrative Values)

**\*\*Finding:\*\*** `qs@6.11.0` in `web-app` via `express@4.18.2`

Factor	Value	Normalized	Weight	Contribution
--------	-------	------------	--------	--------------

-----	-----	-----	-----	-----
-------	-------	-------	-------	-------

CVSS	7.5	0.75	0.30	0.225
------	-----	------	------	-------

KEV	No	0.0	0.35	0.0
-----	----	-----	------	-----

EPSS	0.02	0.02	0.15	0.003
------	------	------	------	-------

Topology	Transitive	0.6	0.10	0.06
----------	------------	-----	------	------

Fix available	Yes (6.11.1)	1.0	0.05	0.05
---------------	--------------	-----	------	------

| Shared across projects | 1 | 0.2 | 0.05 | 0.01 |

| **Total** | | | **0.348** |

**Interpretation:** Medium priority. Not in KEV, low EPSS, but high CVSS. Fix available.

### 10.3 Patch-Order Algorithm

**Input:** Set of findings, each with `package`, `current\_version`, `fixed\_version`,  
`affected\_projects`

**Output:** Ordered list of upgrade suggestions

```
```python
```

```
def suggest_patch_order(findings, max_upgrades=None):
```

```
    """
```

```
    Greedy algorithm: at each step, pick the upgrade that  
    resolves the most findings (weighted by risk score).
```

```
    """
```

```
    remaining = list(findings)
```

```
    upgrades = []
```

```
    while remaining:
```

```
        # For each unique (package, fixed_version), count findings it resolves
```

```
        candidates = {}
```

```
        for f in remaining:
```

```
            key = (f["package"], f["fixed_version"])
```

```

    if key not in candidates:
        candidates[key] = {"findings": [], "total_risk": 0.0}
    candidates[key]["findings"].append(f)
    candidates[key]["total_risk"] += f["risk_score"]

# Pick candidate with highest total risk resolved
best = max(candidates.items(), key=lambda x: x[1]["total_risk"])

upgrades.append({
    "package": best[0][0],
    "target_version": best[0][1],
    "resolves_findings": len(best[1]["findings"]),
    "risk_reduced": best[1]["total_risk"],
    "affected_projects": list(set(
        p for f in best[1]["findings"] for p in f["affected_projects"]
    ))
})

# Remove resolved findings
resolved_ids = {f["id"] for f in best[1]["findings"]}
remaining = [f for f in remaining if f["id"] not in resolved_ids]

if max_upgrades and len(upgrades) >= max_upgrades:
    break

return upgrades

```

...

****Important caveats:****

- This is a ****suggestion****, not a validated fix. MARGVEDHA does not run package managers.
- The algorithm assumes the `fixed\_version` from OSV is compatible; this is not guaranteed.
- If a fix requires a major version change (e.g., `4.x → 5.x`), flag it as "potential breaking change."

10.4 Distinguishing Fix States

State	Description	MARGVEDHA Can Claim
-------	-------------	---------------------

-----	-----	-----
-------	-------	-------

Suggested fix	OSV advisory lists a `fixed` version	"Advisory recommends upgrading to X"
---------------	--------------------------------------	--------------------------------------

Simulated change	MARGVEDHA shows what the graph would look like with X	"If upgraded to X, these findings would be resolved"
------------------	---	--

Resolver-validated	A package manager has confirmed the upgrade satisfies all constraints	**Not claimed**
--------------------	---	------------------------

Test-validated	CI has run tests after the upgrade	**Not claimed**
----------------	------------------------------------	------------------------

11. System Architecture

11.1 Recommended Stack

Component	Technology	Justification
-----------	------------	---------------

|-----|-----|-----|

| Frontend | React + Tailwind CSS + Cytoscape.js | Interactive graph; responsive UI |

| Backend | Python + FastAPI | Fast development; async support; NetworkX integration |

| Graph analysis | NetworkX | Mature; well-documented algorithms |

| Vulnerability data | OSV API | Free; accurate; batch support |

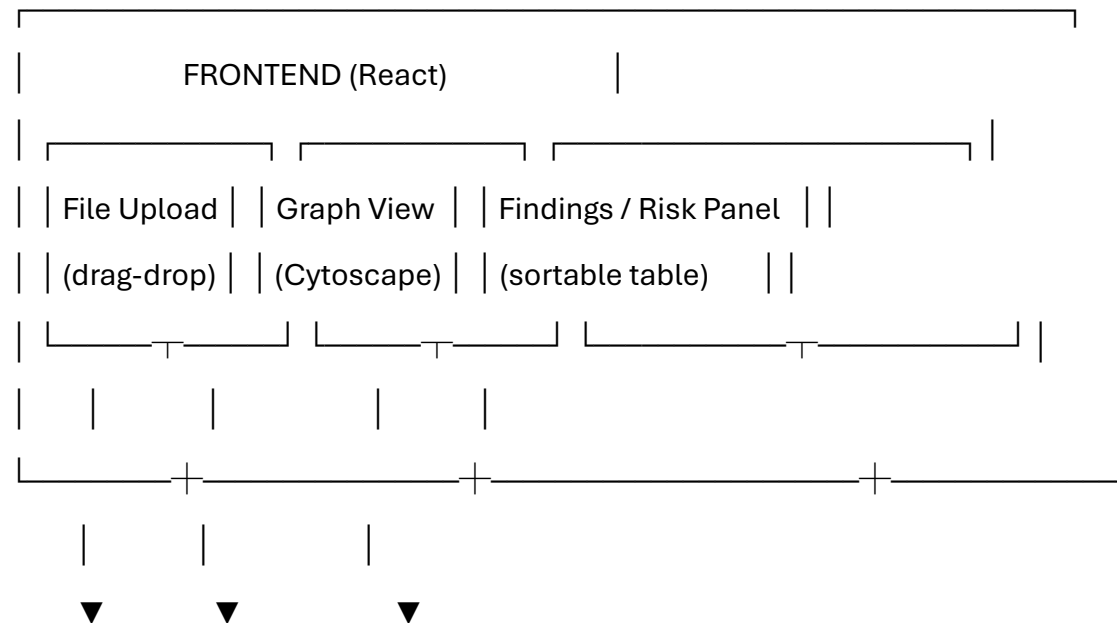
| Persistence | SQLite (MVP) / PostgreSQL (if needed) | SQLite is sufficient for hackathon; no external DB dependency |

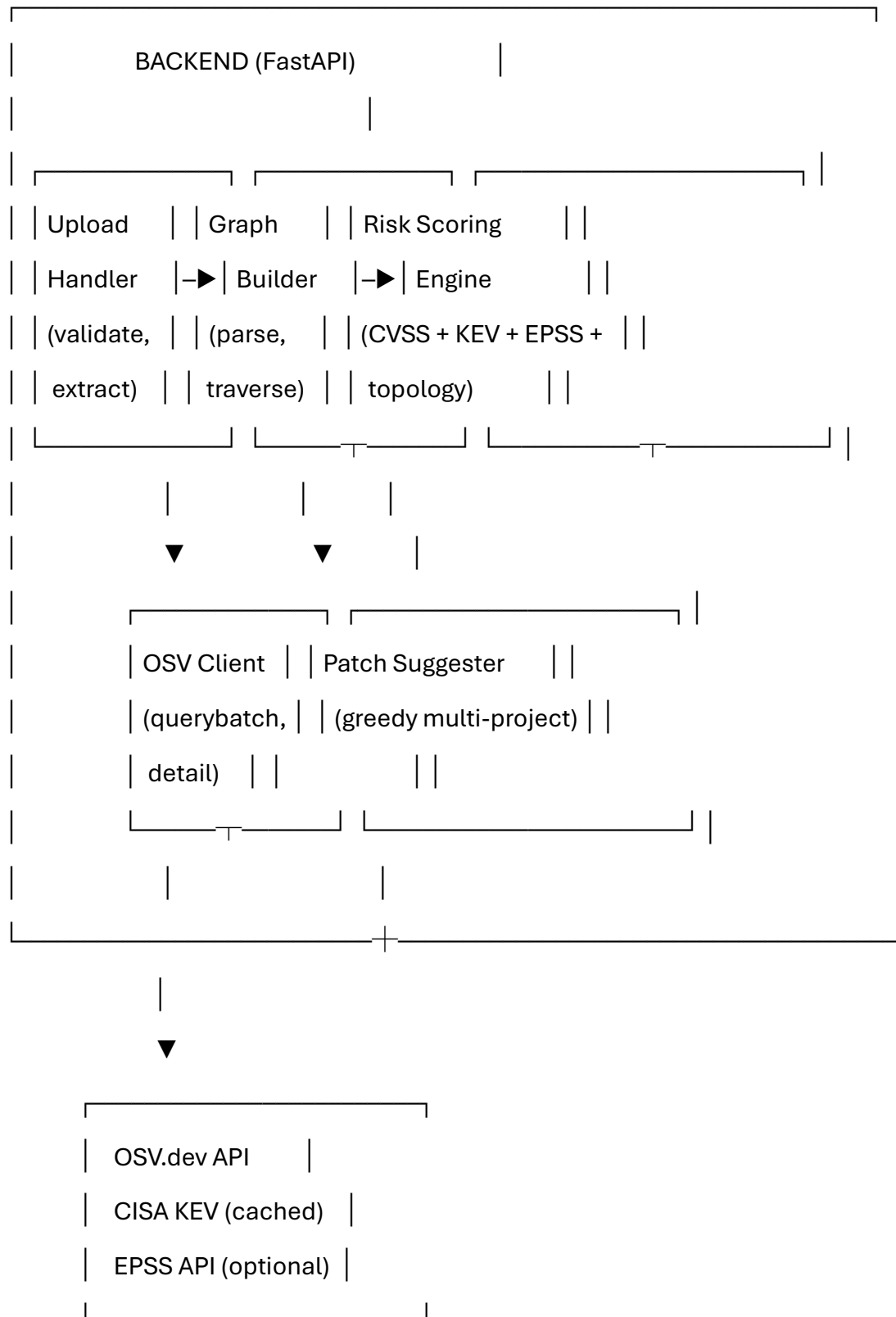
| Deployment | Render / Railway / Fly.io free tier | Student-accessible; supports Python + Node |

****Why not Neo4j?**** For hackathon scale (tens of projects, hundreds of packages), NetworkX in-memory is simpler and sufficient. Neo4j adds operational complexity without proportional benefit.

11.2 Architecture Diagram

...





...

11.3 REST API Endpoints

Method	Endpoint	Purpose	Request Body	Response
POST	/api/v1/upload	Upload manifests/lockfiles	multipart/form-data or JSON with file contents	{upload_id, projects: [...]}
POST	/api/v1/analyze	Analyze uploaded projects	{upload_id} {scan_id}	
GET	/api/v1/scan/{scan_id}	Get scan results	{status, graph, findings, risk_scores}	
POST	/api/v1/simulate	Simulate upgrade impact	{scan_id, upgrades: [{package, target_version}]} {before_graph, after_graph, resolved_findings, new_findings}	
GET	/api/v1/scan/{scan_id}/patch-plan	Get patch suggestions	{package, target_version, resolves_findings, affected_projects}	

11.4 Data Schemas

Project:

``json

```
{
  "id": "proj:web-app",
  "name": "web-app",
  "source_file": "package.json",
  "lockfile": "package-lock.json",
  "ecosystem": "npm",
  "dependency_count": 42
}
```

...

****Finding:****

```json

{

"id": "finding:GHSA-xxxx:lodash:4.17.19",

"vulnerability\_id": "GHSA-xxxx-yyyy-zzzz",

"aliases": ["CVE-2022-24999"],

"package": "lodash",

"ecosystem": "npm",

"version": "4.17.19",

"severity": 7.5,

"fixed\_versions": ["4.17.21"],

"evidence\_state": "PROJECT\_CONTAINS",

"affected\_projects": ["proj:web-app"],

"paths": [

["proj:web-app", "pkg:npm/express@4.18.2", "pkg:npm/qs@6.11.0",

"pkg:npm/lodash@4.17.19"]

],

"risk\_score": 0.75,

"kev": false,

"epss": 0.02

}

...

### ### 11.5 Error Handling

| Error Condition | HTTP Status | Response Body |

|-----|-----|-----|

| Invalid file format | 400 | `{ "error": "unsupported\_file", "detail": "..."} ` |

| OSV API timeout | 503 | `{ "error": "upstream\_unavailable", "detail": "OSV query timed out"} ` |

| No vulnerabilities found | 200 | `{ "findings": [], "message": "No known vulnerabilities"} ` |

| Upload too large | 413 | `{ "error": "file\_too\_large", "max\_size\_mb": 10} ` |

| Malformed manifest | 422 | `{ "error": "parse\_error", "file": "...", "line": N} ` |

### ### 11.6 Caching Strategy

- **OSV responses:** Cache individual vulnerability details for 1 hour (TTL); cache batch query results for 5 minutes
- **KEV catalog:** Cache for 24 hours (updates infrequently)
- **EPSS scores:** Cache for 24 hours (daily updates)
- **Scan results:** Store in SQLite for 24 hours; allow re-fetch by `scan\_id`

### ### 11.7 Deployment Considerations

**Student-accessible hosting:**

- **Render:** Free tier supports Python + Node; spins down after inactivity (acceptable for demo)
- **Railway:** Free tier limited; requires credit card for some features
- **Fly.io:** Free tier supports small VMs; requires CLI familiarity

**\*\*Important caveat:\*\*** Free tiers are subject to change. The team should verify current terms before relying on any provider.

**\*\*Fallback for demo:\*\*** Run locally on the Acer Nitro 5; use ngrok or similar for public access if needed.

## ## 12. Security of the Application

### ### 12.1 Upload-Related Risks

| Risk | Attack Vector | Mitigation |
|------|---------------|------------|
|------|---------------|------------|

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

|                     |                                                   |                                                                                     |
|---------------------|---------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>**Zip Slip**</b> | Malicious ZIP with <code>../`</code> in filenames | Validate all extracted paths; reject absolute paths and <code>..`</code> components |
|---------------------|---------------------------------------------------|-------------------------------------------------------------------------------------|

|                         |                                           |                                                                               |
|-------------------------|-------------------------------------------|-------------------------------------------------------------------------------|
| <b>**Archive bomb**</b> | ZIP with extremely high compression ratio | Limit total uncompressed size; limit file count; enforce per-file size limits |
|-------------------------|-------------------------------------------|-------------------------------------------------------------------------------|

|                           |                                              |                                                                                                   |
|---------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>**Path traversal**</b> | Filename like <code>../../etc/passwd`</code> | Sanitize filenames; use <code>os.path.basename`</code> ; store uploads in isolated temp directory |
|---------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------|

|                                |                                      |                                                                            |
|--------------------------------|--------------------------------------|----------------------------------------------------------------------------|
| <b>**Resource exhaustion**</b> | Huge manifests or thousands of files | Limit upload size (10MB); limit file count (100); limit total lines parsed |
|--------------------------------|--------------------------------------|----------------------------------------------------------------------------|

|                                     |                                                           |                                                     |
|-------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|
| <b>**Arbitrary code execution**</b> | Malicious <code>package.json`</code> with install scripts | <b>**Never execute uploaded code**</b> ; parse only |
|-------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|

|                               |                                     |                                                              |
|-------------------------------|-------------------------------------|--------------------------------------------------------------|
| <b>**Secrets in uploads**</b> | Accidental <code>.env`</code> files | Do not store uploads longer than necessary; warn users in UI |
|-------------------------------|-------------------------------------|--------------------------------------------------------------|

### ### 12.2 Secure Defaults

- **Temp file handling:** Use `tempfile.mkdtemp()` with restricted permissions; delete after scan
- **No code execution:** MARGVEDHA parses manifests; it never runs `npm install`, `pip install`, or `mvn`
- **Network isolation:** If future versions run package-manager commands, sandbox in a network-restricted container
- **Input validation:** Reject binary files; validate JSON/XML structure; limit nesting depth
- **Output sanitization:** Escape package names and version strings in the UI to prevent XSS

### ### 12.3 SSRF Considerations

If MARGVEDHA fetches from user-provided URLs (e.g., a Git repository URL), SSRF is a risk.  
**MVP recommendation:** Do not support URL inputs. Only accept file uploads. This eliminates SSRF entirely.

## ## 13. Dataset and Evaluation Plan

### ### 13.1 Datasets

| Dataset      | URL                                              | License       | Relevance                    |
|--------------|--------------------------------------------------|---------------|------------------------------|
| OSV Database | osv.dev                                          | Open          | Primary vulnerability source |
| CISA KEV     | cisa.gov/known-exploited-vulnerabilities-catalog | Public domain | Exploitation signal          |
| EPSS         | first.org/epss/data_stats                        | Open          | Exploit probability          |

| OSV-Scanner test fixtures | [github.com/google/osv-scanner](https://github.com/google/osv-scanner) | Apache 2.0 | Reference test cases |

| Public repositories with lockfiles | GitHub | Varies | Real-world test data |

### ### 13.2 Test Cases

| Test Case | Input | Expected Behavior | Success Criterion |

|-----|-----|-----|-----|

| Direct vulnerable dependency | `package.json` with `lodash@4.17.19` | Finding with path length 1 | Finding returned |

| Transitive vulnerable dependency | `package-lock.json` with express→qs@6.11.0 | Finding with path length 3 | Finding + path |

| Shared vulnerable package | Two projects both depend on vulnerable lodash | Both projects listed as affected | Affected count = 2 |

| Patched version | `lodash@4.17.21` | No finding | Empty findings |

| No advisory returned | Unknown package | No finding; no crash | Graceful handling |

| Missing version | Manifest with range only, no lockfile | Direct-only analysis; warning | Warning displayed |

| Malformed manifest | Invalid JSON | 400 error; no crash | Error message |

| OSV API failure | Simulated timeout | 503 error; graceful degradation | User informed |

| Unsafe ZIP | Zip with `../` path | Rejected | Upload fails safely |

### ### 13.3 Evaluation Metrics

| Metric | Definition | How to Measure |

|-----|-----|-----|

| Detection precision |  $TP / (TP + FP)$  | Compare findings against ground truth |

| Detection recall |  $TP / (TP + FN)$  | Ensure no known vulnerabilities missed |

| Package-version matching accuracy | Correct matches / total queries | Compare against OSV API results |

| Path correctness | Valid paths / total paths returned | Verify paths exist in lockfile |

| Scan time | Wall-clock seconds | Measure from upload to results |

| Remediation coverage | Findings with available fix / total findings | From OSV `fixed` field |

### ### 13.4 Ground Truth Considerations

**\*\*Critical warning:\*\*** Do not evaluate MARGVEDHA solely against OSV, because OSV is also the detection source. This creates circular evaluation.

**\*\*Recommended approach:\*\***

1. Use OSV for detection (as the tool does)
2. Independently verify a sample of findings manually against GitHub Advisory and NVD
3. Compare graph paths against the lockfile's actual dependency structure
4. For remediation suggestions, verify that the `fixed\_version` actually resolves the advisory (per OSV) and note that resolver validation is not performed

## ## 14. Implementation Roadmap

### ### 14.1 MVP Requirements (Must Have)

1. Upload `package.json` + `package-lock.json` (npm)
2. Parse and build dependency graph
3. Query OSV for package@version

4. Identify vulnerable packages
5. Traverse paths from projects to vulnerabilities
6. Render graph with Cytoscape.js
7. Display findings list with risk scores
8. Show dependency path for each finding

### ### 14.2 Recommended Sequence

| Phase | Task                                 | Duration | Difficulty |
|-------|--------------------------------------|----------|------------|
| ----- | -----                                | -----    | -----      |
| 1     | Manifest parser (npm first)          | 4 hours  | Medium     |
| 2     | OSV API integration                  | 3 hours  | Medium     |
| 3     | Graph builder (NetworkX)             | 4 hours  | High       |
| 4     | Backend API (FastAPI)                | 3 hours  | Medium     |
| 5     | Frontend scaffold (React + Tailwind) | 2 hours  | Low        |
| 6     | Graph visualization (Cytoscape.js)   | 5 hours  | High       |
| 7     | Findings panel + risk scoring        | 4 hours  | Medium     |
| 8     | Path explanation UI                  | 3 hours  | Medium     |
| 9     | Python + Maven parsers               | 5 hours  | High       |
| 10    | Fix-impact simulation                | 4 hours  | High       |
| 11    | Multi-project clustering             | 3 hours  | Medium     |
| 12    | Testing + polish                     | 6 hours  | Medium     |

**\*\*Total: ~46 hours\*\*** (approximately 15 hours per person for a 3-person team)

### ### 14.3 Three-Person Allocation

| Person | Responsibilities |

|-----|-----|

| \*\*Person A (Backend/Data)\*\* | Manifest parsers, OSV client, graph builder, risk scoring, patch suggester |

| \*\*Person B (Frontend/UX)\*\* | React scaffold, Cytoscape.js integration, findings panel, path explanation UI |

| \*\*Person C (Integration/Testing)\*\* | FastAPI endpoints, data schemas, test cases, error handling, deployment |

### ### 14.4 Fallback Plan

| Failure | Fallback |

|-----|-----|

| OSV API unavailable | Use cached OSV snapshot (GCS bucket) loaded into SQLite |

| Cytoscape.js too complex | Fall back to react-flow or a simple tree component |

| Hosting issues | Run locally for demo; use ngrok for public access |

| Maven parsing too hard | Ship npm + PyPI only; document Maven as future work |

### ### 14.5 Features to Defer

- Reachability analysis (requires call-graph construction)
- Automated file modification (risky; OSV-Scanner does this)
- Container scanning
- SBOM generation/consumption (not required by brief)
- User accounts / persistence across sessions

## ## 15. PPT, Report, and Judging

### ### 15.1 PPT Outline (10–12 Slides)

1. **Title:** MARGVEDHA — Trace the Risk. Secure the Path.
2. **Problem:** Supply-chain vulnerability propagation is opaque to developers
3. **Gap:** Existing tools show findings; few explain paths or simulate fixes
4. **Solution:** Multi-project dependency graph with explainable paths and fix simulation
5. **Architecture:** React + FastAPI + NetworkX + OSV
6. **Demo (screenshot/video):** Upload, graph view, path explanation, fix simulation
7. **Data Sources:** OSV, CISA KEV, EPSS
8. **Risk Model:** Multi-factor scoring with explicit evidence tiers
9. **Evaluation:** Test cases, metrics, limitations
10. **Competitive Positioning:** What exists, what MARGVEDHA adds
11. **Limitations:** No reachability; manifest-only inputs limited
12. **Conclusion:** Evidence-based triage for developer workflows

### ### 15.2 Judge Questions with Defensible Answers

**Q1:** Why is MARGVEDHA necessary when OSV-Scanner, Snyk, and Dependabot exist?

**A:** Those tools excel at detection and enterprise workflows. MARGVEDHA focuses on **developer-facing triage for multi-project scenarios**: it shows the precise dependency path for each finding, simulates the impact of candidate upgrades across projects, and clusters shared vulnerable dependencies. It's not a replacement; it's a complementary analysis layer.

**\*\*Q2: How is this different from OSV-Scanner's guided remediation?\*\***

**\*\*A:\*\*** OSV-Scanner's guided remediation modifies files for a single project. MARGVEDHA does not modify files; it simulates what would happen across multiple projects if an upgrade were applied. The multi-project clustering and before/after graph comparison are the differentiators.

**\*\*Q3: Do you perform reachability analysis?\*\***

**\*\*A:\*\*** No. We explicitly do not claim runtime reachability. We report **\*\*path reachability\*\*** — a structural dependency path exists — and label every finding with its evidence state. Reachability analysis requires language-specific call-graph construction (as Snyk does for Java) and is deferred.

**\*\*Q4: How do you handle missing lockfiles?\*\***

**\*\*A:\*\*** We report that transitive analysis is limited. If only a manifest is provided, we analyze direct dependencies and clearly label the limitation. We never claim complete transitive coverage without a lockfile.

**\*\*Q5: How is risk score calculated?\*\***

**\*\*A:\*\*** Multi-factor formula combining CVSS severity, KEV membership, EPSS probability (if available), topology (direct/transitive), fix availability, and shared-dependency count. Weights are configurable and the formula is exposed to the user.

**\*\*Q6: What about false positives?\*\***

**\*\*A:\*\*** We distinguish three evidence states: advisory match, project includes package, and path-reachable. We never conflate these. We also report when OSV returns no findings (which could be a false negative due to advisory lag).

**\*\*Q7:** How do you validate the patch suggestions?

**\*\*A:\*\*** We do not validate. Our suggestions are advisory-based (OSV `fixed` version) and simulated (graph after hypothetical upgrade). We label them as "suggested fixes" and flag major version changes as potential breaking changes.

**\*\*Q8:** What is the graph algorithm used?

**\*\*A:\*\*** NetworkX `all\_shortest\_paths` for path enumeration; BFS for affected-project traversal. We do not use centrality as a risk signal because topology does not prove exploitability.

**\*\*Q9:** How do you handle multiple versions of the same package?

**\*\*A:\*\*** Each package@version is a distinct node. A project can depend on multiple versions via different paths. We report all affected versions separately.

**\*\*Q10:** What about Python and Maven?

**\*\*A:\*\*** MVP prioritizes npm (full lockfile support). PyPI lockfiles (poetry.lock, Pipfile.lock) are supported in Phase 2. Maven is limited to direct dependencies from pom.xml; transitive analysis requires Maven execution, which we do not perform.

**\*\*Q11:** How do you ensure the tool itself is secure?

**\*\*A:\*\*** No code execution. Static parsing only. Upload validation (size, file count, path traversal prevention). Temp files cleaned after scan. No URL inputs (eliminates SSRF).

**\*\*Q12:** What is the novelty contribution?

**\*\*A:\*\*** The integration of multi-project batch analysis, explainable path rendering, fix-impact simulation, and evidence-tiered labeling in a single free tool. Each component exists elsewhere, but this specific integration for developer triage is not available in free tools.

**\*\*Q13:** How do you evaluate accuracy?

**\*\*A:\*\*** Test cases with known vulnerable/patched versions; comparison against OSV API results; manual verification of a sample against GitHub Advisory; path validation against lockfile structure.

**\*\*Q14:** What happens if OSV API fails?

**\*\*A:\*\*** We return a 503 error with a clear message. For MVP, we do not maintain a local mirror. Future versions could use the OSV GCS snapshot.

**\*\*Q15:** How does this help a developer who isn't a security expert?

**\*\*A:\*\*** The graph view shows the dependency chain in familiar terms. The risk score is explained, not opaque. The patch plan tells them what to upgrade first and why. The evidence labels prevent them from panicking over unreachable code.

**## 16. References**

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## ## 17. Final Decision Memo

### \*\*First features to build:\*\*

1. npm manifest + lockfile parsing
2. OSV batch query integration
3. Dependency graph construction (NetworkX)
4. Path traversal and affected-project identification
5. Graph visualization with Cytoscape.js
6. Findings panel with evidence labels

**\*\*Primary proposed contribution:\*\***

Multi-project explainable propagation graph with fix-impact simulation.

**\*\*Features to defer:\*\***

- Reachability analysis
- Maven transitive analysis
- SBOM generation
- Automated file modification
- Container scanning
- User accounts

**\*\*Most important technical risks:\*\***

1. Manifest-only inputs produce incomplete graphs — must label clearly
2. OSV API rate limits/downtime — cache aggressively
3. Cytoscape.js performance with large graphs — test early
4. Graph traversal bugs (cycles, duplicate paths) — unit test extensively

**\*\*Evidence needed to validate innovation:\*\***

1. Demonstrate multi-project shared-dependency clustering on real test data
2. Show fix-impact simulation reducing findings count
3. Verify path correctness against lockfile structure
4. Compare risk ranking against CVSS-only baseline

**\*\*Claims safe to make:\*\***

- "MARGVEDHA maps transitive vulnerability propagation with explainable paths."

- "MARGVEDHA simulates the impact of candidate upgrades across multiple projects."
- "MARGVEDHA labels findings with evidence tiers, distinguishing advisory matches from path reachability."

**\*\*Claims not yet supported:\*\***

- "MARGVEDHA reduces false positives through reachability analysis."
- "MARGVEDHA is the first tool to combine graph analysis with remediation planning."
- "MARGVEDHA validates that suggested fixes are compatible."